

# Decision framing, evidence boundaries, tool readiness

Quantitative Methods for Business Analysis. Complete explanations, worked examples, practice and transfer cases for every idea in the session.

≈ 54 min read

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Concise edition

## 1 · READ

Open a section, read its central idea, then its worked example.

## 2 · APPLY

Try the guided practice and the lab task linked in each section.

## 3 · CHECK

Answer the study check in the writing space without looking back.

## Your study route

### 1.1

#### Quantitative Analysis as a Decision Discipline

5 topics · Part A

### 1.2

#### From a Managerial Concern to a Bounded Analytic Question

5 topics · Part E

### 1.3

#### Testable Claims, Assumptions, and Evidence Boundaries

5 topics · Part D

### 1.4

#### Minimum Viable Readiness in Excel and SPSS

5 topics · Part B · Part C · Part F

## ▼ What you will be able to do

- Explain why quantitative business analysis begins with a stakeholder decision and an evidence question rather than with software or computation.
- Distinguish a business problem, a stakeholder decision, an analytic question and a software task.
- Construct a question frame that identifies the stakeholder, decision, scope, time horizon, observable outcome, reference point and evidence need.
- Audit a proposed claim for testability and revise language that exceeds what the anticipated evidence could support.

- Demonstrate minimum readiness in Excel and SPSS by opening or importing provided files, executing provided steps, inspecting results, saving and reopening artifacts, and recording provenance.
- Classify a request as ready to analyze, clarification required, evidence blocked or tool blocked, and justify the classification.

### ▼ Before you start

Read the PDF edition for study, annotation or printing; the HTML, notebook and Markdown editions are also available. A running kernel, software installation and an account connection are not required to read this toolkit. Keep notes in the response spaces or in your own study document.

The code blocks and formulas are complete teaching references. The worked results and verification directions are reproduced from the session materials and the S01P practice guide; this toolkit does not execute code or independently establish those results. If you choose to try a reference, use the inputs, tools and conditions specified beside it.

Session 01 practice is ungraded. Nothing is submitted or uploaded.

Before reading, write one idea you already understand and one question you want the session to answer.

Starting notes

My prior understanding and learning question:

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# 1.1 · Quantitative Analysis as a Decision Discipline

## IN CLASS

### [Session slides](#)

The eight-step workflow 13

Missing leading zeros 34

## PRACTISE IT IN THE LAB

- [Part A: What does one row of the order records represent?](#) · 8 min

About 15 min read · PDF pages 4–16 · [Short version \(Concise\)](#)

## Learning goals

- Explain why quantitative business analysis begins with a stakeholder decision and an evidence question rather than with software or computation.
- Distinguish a business problem, a stakeholder decision, an analytic question and a software task.
- Separate what evidence can contribute to a decision from what remains managerial judgment.

## The Four Levels of an Analytical Request

### CONCEPT What the Four Levels Mean

This course treats quantitative analysis as a decision discipline, not a set of software procedures. Data creates value only when evidence reduces uncertainty around a consequential managerial choice. Requests often blend a symptom, a preferred solution and a software command into one sentence, such as "Open SPSS, analyze our support tickets, and stop customer churn." Before loading any file, separate the sentence into four levels. A **business problem** is an observed condition that needs attention, such as rising cancellations. A **stakeholder decision** is a commitment of money, staff or policy owned by an accountable leader. An **analytic question** names the evidence needed to reduce uncertainty about that decision. A **software task** is a mechanical operation in a tool such as Excel or SPSS. Mixing the levels is how an accurate calculation ends up answering the wrong question.

## CONCEPT How the Evidence-to-Decision Chain Works

The four levels sit inside a longer chain: a consequential condition, an accountable stakeholder, a pending decision, the relevant uncertainty, an evidence question, the quantitative analysis, a bounded interpretation and, finally, human judgment. The decision determines which uncertainty matters, and the uncertainty determines which evidence is relevant. The software task appears only inside the analysis link, late in the chain. Opening a tool before the earlier links are clear is the **tool-first trap**. The chain ends with judgment, not with the result: evidence informs a decision but does not take responsibility for it. Ask every requester: "What specific managerial decision will change depending on the outcome of this calculation?"

## WORKED EXAMPLE Worked Example: Decomposing the OmniRetail Request

OmniRetail is the session's simulated case: a subscription retailer with 2.4 million active accounts. Leadership observed a 14 percent quarterly increase in cancellations. The Vice President of Customer Operations proposed hiring 40 weekend support representatives, an annual commitment of 450,000 dollars, pointing to a Monday backlog of 4,200 unresolved tickets. Separating the raw request gives the four levels below.

**WORKED EVIDENCE**      **Population:** OmniRetail, a simulated subscription retailer with 2.4 million active accounts

**Period:** The quarter in which cancellations rose 14 percent

**Unit:** One level of the analytical request

| Level                   | OmniRetail example                                                                            | What it needs next            |
|-------------------------|-----------------------------------------------------------------------------------------------|-------------------------------|
| 1. Business problem     | Quarterly cancellations rose 14 percent                                                       | A named owner                 |
| 2. Stakeholder decision | Authorize 450,000 dollars per year for 40 weekend representatives                             | Options and a deadline        |
| 3. Analytic question    | Do subscribers with unresolved weekend tickets cancel more often than comparable subscribers? | Observable, approved evidence |
| 4. Software task        | Filter and summarize ticket and cancellation records in SPSS or Excel                         | Correct execution             |

**Interpretation:** Only Level 2 commits budget or policy. The software task is specified last, not first.

### **CONCEPT** Interpreting the Decomposition

Once separated, the request shows what is missing. Analysis cannot "stop churn"; it can measure associations and inform a retention strategy. SPSS was named before anyone defined a population, comparison or time horizon. The Vice President owns the decision; the analyst formulates the question with that leader; a technically perfect software task is still useless if it summarizes the wrong records.

### **⚠️ COMMON MISTAKE** Common Misconception: Opening Software Is the Start of Analysis

Many people believe analysis begins when the file opens. Opening a workbook and running a provided step is a software task, not an evidence-to-decision analysis. A related belief is that a larger dataset is more credible. More records do not help if the data are irrelevant, poorly documented or mismatched to the decision, and uncurated data can add noise and misleading patterns. The correction is the same each time: name the stakeholder and decision before the tool or the file.

### **🚧 LIMITS AND CONDITIONS** Limits of the Four-Level Classification

A request may be an unresolved mixture; separate it, and write one frame per materially different decision. Several stakeholders may be involved; identify the decision owner and record the others whose constraints matter. If a manager cannot name a decision, ask what action, commitment or allocation could change after the analysis. Finally, some important requests are not ready for quantitative analysis. Analysis fits when observable evidence can materially reduce uncertainty about a meaningful decision. When evidence is inaccessible, prohibited, irrelevant or likely to mislead, pausing is the professional response.

### **🎯 GUIDED PRACTICE** Guided Practice: Classifying Four Requests

The session opened with four requests shown before any definitions. Task 1: Classify each as a business problem, stakeholder decision, analytic question or software task. Task 2: For the decision, name the accountable role and time horizon. Check your answers in the Reference summary for this section.

## WORKED EVIDENCE

**Population:** Opening classification requests from Section 1

**Period:** Session 01 opening activity

**Unit:** One request sentence

| Request                                                                                                  | My classification |
|----------------------------------------------------------------------------------------------------------|-------------------|
| Quarterly cancellations have risen.                                                                      |                   |
| Should the service vice president add weekend coverage next month?                                       |                   |
| What approved evidence would reduce uncertainty about whether weekend demand warrants a coverage change? |                   |
| Open the workbook and run the provided step.                                                             |                   |

**Interpretation:** Each sentence belongs to one level.

**Optional independent practice: my response**

*Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.*



TRANSFER

## Changed-Case Transfer: Package Delivery Delays

A logistics company faces package delivery delays, and customer complaints are rising. The Chief Operating Officer is considering whether to authorize 200,000 dollars to lease ten additional delivery vans. An analyst is asked to "filter the dispatch logs and calculate average delay times by route in Excel." Task 1: Place each element into one of the four levels. Task 2: Write an analytic question that could show whether delays come from driver shortages or from warehouse dispatch bottlenecks. Task 3: Explain why leasing vans is a decision rather than a problem. Task 4: Which level commits budget or policy?

**Optional independent practice: my response**

*Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.*



#### KEY TERMS

### Key terms

**Business Problem:**

An observed organizational condition or performance gap requiring managerial attention.

**Stakeholder Decision:**

A specific commitment of capital, staffing or policy owned by an accountable leader or role.

**Analytic Question:**

The neutral question that identifies what evidence is needed to reduce uncertainty about the decision.

**Software Task:**

A mechanical procedure executed in an application such as Excel or SPSS.

**Tool-First Trap:**

Opening software before the decision owner, pending action and evidence boundary are established.



#### STUDY CHECK

### Put the learning into action

- Classify any incoming request as a problem, decision, question, task or unresolved mixture before opening a tool.
- Extract the implied stakeholder, decision and evidence need from a short managerial sentence.
- State the missing information to clarify first, starting with the decision that could change.
- Reflect on whether your habits begin with a preferred tool, an available file or a stakeholder decision.

# The Eight-Step Workflow and the Boundary Between Evidence and Judgment

## STEP BY STEP What the Eight-Step Workflow Means

The course uses eight steps as the spine of every analysis: **Frame, Inspect, Choose, Assume, Analyze, Quantify Uncertainty, Recommend and Communicate**. Inspect audits the data's **provenance** (where it came from), its **grain** (what one row represents), missing values and baseline quality. Choose and Assume select a valid method and record the conditions it needs. Analyze runs transparent, reproducible calculations. Quantify Uncertainty and Recommend state how uncertain the result is and derive a bounded action. Communicate turns the findings into a defensible recommendation. The guided demonstration walks the same logic in plainer words, from identifying the issue to communicating the boundary.

## CONCEPT How the Steps Act as Quality Gates

Each step protects the next. If Step 1 misframes the decision, Step 2 inspects the wrong population. For OmniRetail, the team must first check whether each support-log record is a unique customer or an individual ticket, because that grain changes every later count. Assumptions are written before analysis so criteria cannot be bent to fit results, and communication comes last so it reports the completed chain.

## WORKED EXAMPLE Worked Example: Repairing a Tool-First Request

A manager writes: "Run SPSS on every ticket and stop churn." The analyst finds four defects: a tool named before the problem, an impossible guarantee, no decision owner, and no population, comparison or time horizon. The repair fixes each. The reframed question reads: "Among active monthly subscribers, is an unresolved weekend support ticket associated with a meaningful increase in 60-day cancellation rates compared to weekday tickets?" The fixed 60-day window matters: without it, the definition of churn drifts and results cannot be reconciled across tools.

**WORKED EVIDENCE**      **Population:** Active United States monthly subscribers at OmniRetail (simulated case)

**Period:** Trailing twelve months, with a 60-day cancellation window after each ticket

**Unit:** Not yet confirmed; ticket or customer is settled at Step 2, Inspect

| Defect in the original request       | Repair in the reframed question                                                     |
|--------------------------------------|-------------------------------------------------------------------------------------|
| Names SPSS before the problem        | Starts from the 450,000-dollar staffing decision                                    |
| Promises to "stop churn"             | Asks whether a pattern is associated with cancellation                              |
| No decision owner                    | Vice President of Customer Operations                                               |
| No population, comparison or horizon | US monthly subscribers; unresolved weekend versus resolved weekday tickets; 60 days |

**Interpretation:** The reframe keeps executive intent but makes a question that evidence could answer either way.

### CONCEPT Interpreting Evidence Against Managerial Judgment

**Quantitative evidence** (patterns, estimates, distributions and uncertainty bounds) reduces uncertainty. **Managerial judgment** belongs to the accountable executive, who combines evidence with strategy, budget, feasibility, legal requirements and ethics. The slides give an illustration: even if customers with weekend ticket delays showed a 6 percent higher cancellation rate, leaders would still need to weigh 450,000 dollars a year against alternatives such as self-service knowledge bases or chatbot triage. The guided demonstration's table makes the same split. Evidence may contribute demand patterns in approved records, missingness and unusual observations, and consistent summaries across tools. Judgment still covers budget and staffing tolerance, contractual and operational constraints, and which tradeoffs to accept.

### ⚠️ COMMON MISTAKE Common Misconception: A Defensible Finding Is a Mandate to Act

It is tempting to treat a statistically defensible finding as an instruction. The session states that statistical significance does not equal mandatory action. Constraints may make an intervention uneconomical, and leaders may reasonably reject a sound recommendation because of cost, bottlenecks or other priorities. Data never makes a decision; accountable people do.



## STEP BY STEP

### Limits of the Workflow and of Evidence

The workflow requires that a method be chosen and justified, but it does not choose one for you. Session 01 practices only Frame and Inspect, plus tool readiness; methods come in later sessions. Evidence has limits too. Service records might show when demand occurred, but they cannot determine budget tolerance, staff availability, contracts or whether weekend coverage is the preferred response. When the decision owner or intended use is unknown, the workflow stops at Frame until clarification arrives.



## GUIDED PRACTICE

### Guided Practice: The Monday Backlog Dilemma

OmniRetail runs support Monday through Friday. Every Monday, managers find 4,200 unresolved tickets and assume the backlog shows weekend frustration that fuels cancellations. Plausibility is not proof: a Monday queue does not show that the tickets were submitted over the weekend. Task 1: In 90 seconds, write three data facts to verify in the raw ticket logs before supporting the budget request. Task 2: Write one plausible alternative explanation for the Monday surge. Task 3: Name the most important timestamp to inspect, and explain why queue-entry time is not enough.

Optional independent practice: my response

*Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.*



## TRANSFER

### Changed-Case Transfer: Weekend Coverage for Service Operations

In the guided demonstration, the Vice President of Service Operations writes that quarterly service cancellations have risen and she is considering whether weekend coverage should change next month. She asks the analysts to clarify the decision question, inspect the approved service records, and say what the evidence can support and what still needs judgment. Task 1: Identify the stakeholder, decision and time horizon. Task 2: Write a neutral evidence question that could support or challenge a coverage change. Task 3: Complete: "The evidence can show , ***but the decision maker must still judge*** ."

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.

## CONCEPT Python Inspection of the Shared Workbook (Optional Colab Reference)

Code reference for reading and optional independent practice.

```
import pandas as pd

FILE = "/content/Session 01 - Analysis Practice Workbook.xlsx"
data = pd.read_excel(FILE, sheet_name="Data")

print("Rows and columns:", data.shape)
print("\nColumn names:")
print(data.columns.tolist())
print("\nMissing values by column:")
print(data.isna().sum())
print("\nFirst five rows:")
display(data.head())
```

### DEFINITIONS

- `pd.read_excel(..., sheet_name="Data")`: Reads the Data sheet, the single source shared by Excel, SPSS, Python and R.
- `data.shape`: Reports the number of rows and columns.
- `data.isna().sum()`: Counts blank cells in each column.
- `display(data.head())`: Shows the first five rows, so you can see what one row looks like.

### INPUTS AND ASSUMPTIONS

The practice workbook uploaded through the Files panel of a Python Google Colab notebook. Python is optional and ungraded.

### Worked steps

- 1 Step 1: Upload the workbook and run the cell once.
- 2 Step 2: Read the dimensions, column names and missing counts before looking at any statistic.
- 3 Step 3: Compare them with Start Here and the S01P guide, which describe 120 order records.

|                                   |                                                                                                                                                                                       |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WORKED RESULT</b>              | 120 rows and 12 columns, from order_id to customer_rating, with 2 blank delivery_minutes values and 1 blank order_value value.                                                        |
| <b>WHAT THE RESULT MEANS</b>      | This is Step 2, Inspect. It shows size, fields and missingness without making any decision. If the numbers differ from the guide, stop and check that every tool reads the same file. |
| <b>WHEN TO USE IT</b>             | As an optional cross-check of the shared source.                                                                                                                                      |
| <b>LIMITS AND CONDITIONS</b>      | It does not replace the Excel and SPSS practice, and a matching row count does not prove the values are accurate.                                                                     |
| <b>HOW TO CHECK THE REFERENCE</b> | The missing counts should equal the Missing N values in Part B of the practice: 2, 1 and 0.                                                                                           |



#### KEY TERMS

#### Key terms

##### **Eight-Step Workflow:**

Frame, Inspect, Choose, Assume, Analyze, Quantify Uncertainty, Recommend and Communicate.

##### **Grain:**

What one row of a table represents, such as one order, one ticket or one customer.

##### **Provenance:**

The record of where data came from and how it was handled.

##### **Quantitative Evidence:**

Measurable patterns, estimates and uncertainty bounds whose role is to reduce uncertainty.

##### **Managerial Judgment:**

The decision maker's integration of evidence with strategy, budget, feasibility, legal constraints and ethics.



#### STUDY CHECK

#### Put the learning into action

- Use the eight steps as headings in your working notes.

- Rewrite a tool-first instruction as a question that names the owner, population, comparison and time horizon.
- Complete "The evidence can show , *but the decision maker must still judge* " for each request.
- Reflect on how pressure to appear decisive can encourage overstated conclusions.



#### REFERENCE SUMMARY

#### Reference summary

Use these concise reminders after reading the complete explanations above.

#### CONCEPT

#### The Four Levels

##### NAME EACH LEVEL

- Problem: an observed condition. Decision: a commitment owned by a leader. Question: the evidence needed. Task: a tool operation.

##### CHECK THE OPENING CLASSIFICATION

- "Quarterly cancellations have risen" is a problem. "Should the service vice president add weekend coverage next month" is a decision. "What approved evidence would reduce uncertainty..." is an analytic question. "Open the workbook and run the provided step" is a software task.



#### GO DEEPER

#### Why Start with the Decision?

##### THE DECISION SETS RELEVANCE

- It determines which uncertainty matters and therefore which evidence is relevant.

#### CONCEPT

#### The Eight Steps

##### RECITE THE SEQUENCE

- Frame, Inspect, Choose, Assume, Analyze, Quantify Uncertainty, Recommend, Communicate.

## CONCEPT Evidence and Judgment

Evidence reduces uncertainty; judgment decides

- Budget, staffing, contracts, feasibility, ethics and tradeoffs stay with the decision owner.

SAY IT IN ONE  
SENTENCE

- "The evidence can show , *but the decision maker must still judge* ."

## ✓ STUDY CHECK Study check: Quantitative Analysis as a Decision Discipline

Use this ungraded check after reading. Close the explanation before responding, then reopen it to identify a specific correction.

1. Explain one central idea in your own words and connect it to a worked example above.
2. Reconstruct the example's inputs, sequence and interpretation; state what evidence supports each step.
3. Change one assumption or condition. Explain whether the same approach still applies.
4. Name a likely mistake and the check you would use to detect it.

My response

*Write your explanation, evidence and changed-condition reasoning here.*

MY READINESS

Choose one: **Can explain independently** / **Can explain with the reference**  
/ **Need to revisit.**

*Record the passage to revisit and the next action you will take.*

Carry one unresolved question into the next section.

## ✍ CONNECT TO THE LAB Connect this section to the lab

Open Part A in the lab guide →

Session lab reference

S01P (ungraded practice): MetroFresh Logistics: Are our order records ready to use as evidence?

### Open the assigned task

Part A: What does one row of the order records represent?

### Student lab file

S01P\_Student\_Lab.html , with Session 01 - Analysis Practice Workbook.xlsx and Session 01 - SPSS Practice Syntax.sps

### Workbook location

'Data Audit'!B6:B11 and 'Data Audit'!B15:D19

### SPSS location

GET DATA block and VARIABLE LEVEL command in Session 01 - SPSS Practice Syntax.sps

### Primary task

Apply Step 2, Inspect: state what one row represents, which field identifies it, and whether a repeated customer\_id is an error or expected.

### Supporting task

Check that items\_unfulfilled never exceeds items\_ordered, and give a measurement level, reason and treatment for five fields.

### Excel task

In 'Data Audit'!B9, count the table rows with the formula on the Part A card (check value 120). Write short responses in B6:B8 and B10:B11 and complete B15:D19. Never edit the Data sheet.

### SPSS task

Run the GET DATA block, confirm 120 cases, and compare the levels in Variable View with your Part A choices.

### Bring back to the discussion

One sentence that says what one row represents.

This session's practice is ungraded and nothing is submitted. Follow the practice guide's own setup steps, instructions and checklist. The Toolkit is a study reference and does not change those requirements.

### My connection

*Which idea from this section supports my lab decision, and what evidence will I check?*

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## 1.2 · From a Managerial Concern to a Bounded Analytic Question

### IN CLASS

#### [Session slides](#)

Evidence and managerial judgment 14

Causal claims and observational data 22

### PRACTISE IT IN THE LAB

- [Part E: Can we use these records as evidence, or should we pause?](#) · 10 min

About 13 min read · PDF pages 17–28 · [Short version \(Concise\)](#)

### Learning goals

- Construct a question frame that identifies the stakeholder, decision, scope, time horizon, observable outcome, reference point and evidence need.
- Replace vague terms such as better, successful, recent and customers with context-specific language.
- Narrow or separate a request that combines several decisions, and recognize scope drift before it stalls a decision.

## Anatomy of a Bounded Question

### CONCEPT What a Bounded Question Means

A **bounded analytic question** states enough context, scope and time to limit what the answer represents. Its elements are the stakeholder, the pending decision, the business context, the included and excluded **scope**, the **time horizon**, the observable outcome, a useful **reference point** and the evidence need. The decision need not be yes or no; it may be choosing, prioritizing, allocating, revising or pausing. The reference point is a comparison, such as a prior period or another group, that makes the evidence interpretable. Without boundaries, two analysts using the same data can reach conflicting conclusions because they filtered differently.

## CONCEPT How the Components Fit Together: Three Tiers

The slides group the components into three tiers. **Decision Context** names the accountable leader, the pending choice and the minimum actionable difference, anchoring the question to a real capital, staffing or policy commitment. **Evidence Scope** names the population and period, including units, geography and time horizon, plus explicit exclusions. **Measurement Plan** names the observable outcome, the comparison baseline and a verified data source, with an audit trail confirming provenance before data are touched. In this session you name the outcome and reference point without building detailed measures; metrics, variables and grain are the next session's work.

## WORKED EXAMPLE Worked Example: The OmniRetail Bounded Frame

Fill every component for the OmniRetail staffing request. The minimum actionable difference, agreed before any data are examined, is five **percentage points**. A percentage point is the arithmetic difference between two percentages: 10 percent to 15 percent is five percentage points. Compare the vague question "Do weekend tickets hurt retention?" with the bounded one: "Among active United States monthly consumer subscribers over the trailing twelve months, is an unresolved weekend support ticket associated with a higher 60-day cancellation rate compared to resolved weekday tickets?"

**WORKED EVIDENCE**      **Population:** Active United States monthly subscribers at OmniRetail, enterprise accounts excluded (simulated case)

**Period:** Trailing twelve months, with a 60-day observation window after each ticket

**Unit:** One component of the question frame

| Tier             | Component                     | OmniRetail entry                                        |
|------------------|-------------------------------|---------------------------------------------------------|
| Decision Context | Stakeholder                   | Vice President of Customer Operations                   |
| Decision Context | Pending choice                | Allocate 450,000 dollars per year for weekend support   |
| Decision Context | Minimum actionable difference | At least five percentage points higher cancellation     |
| Evidence Scope   | Population                    | Active United States monthly subscribers                |
| Evidence Scope   | Exclusion                     | Enterprise accounts                                     |
| Evidence Scope   | Time horizon                  | Trailing twelve months; 60-day outcome window           |
| Measurement Plan | Outcome                       | 60-day cancellation rate                                |
| Measurement Plan | Reference point               | Resolved weekday tickets                                |
| Measurement Plan | Data source                   | Support ticket and cancellation records, to be verified |

**Interpretation:** Every cell is filled before any record is extracted. An empty cell is a clarification question, not a detail for later.

### CONCEPT Interpreting a Bounded Frame

A completed frame shows whose decision the evidence serves, what part of the business it represents and what observation would be informative. It also exposes the analyst's own choices. Boundaries you add yourself, such as a provisional time window, should be labeled as assumptions. The reference point is essential: without a comparison group, you cannot say whether an observed rate is elevated.

 **COMMON MISTAKE**

## **Common Misconception: Available Data Should Determine the Question**

It is tempting to ask whatever the available data can answer. The session reverses the order: data affect feasibility, but the decision determines relevance. A question can be answerable yet irrelevant; counting tickets by weekday does not show whether unresolved weekend tickets relate to cancellation. A broad question is also not automatically more strategic; breadth can combine incompatible decisions.



**LIMITS AND CONDITIONS**

## **Limits of Bounding**

Every exclusion narrows what the finding covers. Excluding enterprise accounts makes the OmniRetail comparison cleaner, and it also means the result cannot justify enterprise support policy. The session frames the tradeoff as a question broad enough to matter and narrow enough to answer honestly. Bounding cannot create missing evidence: if the preferred source is unavailable, document it, negotiate an approved proxy or revise the scope. Ask whose experience the scope might leave out.



**GUIDED PRACTICE**

## **Guided Practice: Framing "Improve Customer Retention"**

The session posed a three-word request: "Improve customer retention." Work silently first: underline every vague term and draft one question. Task 1: Name an accountable role and a decision it could face. Task 2: Bound the population, enrollment period and observation horizon. Task 3: State the observable outcome and reference point without building a detailed metric. Task 4: List at least two excluded contexts. Task 5: Compare with the session's modeled frame: a customer success director deciding whether to commission a limited onboarding pilot asks what approved account records for United States monthly subscribers enrolled from January through March show about documented continuation 60 days after enrollment, relative to an agreed prior-period reference; annual plans, other countries and behavior after 60 days are outside the frame.

### **Optional independent practice: my response**

*Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.*

**TRANSFER****Changed-Case Transfer: The Same Concern, a Different Decision**

Keep the concern, customer retention, but change the stakeholder: a marketing leader is deciding whether to revise acquisition messaging. Task 1: Write a bounded question for this leader. Task 2: Compare it with the customer success director's frame and name at least two components that change. Task 3: Explain why both framings can be valid.

**Optional independent practice: my response**

*Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.*

**KEY TERMS****Key terms****Bounded Analytic Question:**

A question that states the stakeholder, decision, context, scope, time horizon, observable outcome, reference point and evidence need.

**Scope:**

The people, operations, locations, conditions and periods represented, stated together with what is excluded.

**Exclusion Criterion:**

An explicit rule, recorded in advance, stating which cases are outside the question.

**Time Horizon:**

The fixed period and observation window for outcomes, such as 60 days after a ticket.

**Reference Point:**

A comparison baseline that makes evidence interpretable.

**STUDY CHECK****Put the learning into action**

- Fill all three tiers before extracting any data.
- State one included and one excluded context for every question.
- Label boundaries you set without stakeholder confirmation as assumptions.

- Reflect on whether you keep a frame because it is easy to answer rather than relevant.

## Scope Discipline, Vague Terms, and Clarification

### CONCEPT What Scope Discipline Means

Scope discipline keeps a question inside its agreed boundaries and changes them only on purpose. It starts with language: "recent" could mean a week or a year, and "customers" may or may not include enterprise accounts. Replace each vague term with bounded language that could later be measured. The session names three scoping mistakes. **Uncontrolled scope drift:** stakeholders add questions, cohorts and metrics without adjusting timelines. **Premature metric coding:** analysts write formulas before teams agree what terms such as "active subscriber" or "resolution time" mean. **Concealed limitations:** known gaps or exclusions are hidden, so leaders assume findings apply universally.

### CONCEPT How Clarification Works: The Clarification Ladder

When framing information is missing, climb the clarification ladder. First ask what decision could change. Then ask who owns it and when it must be made. Then ask which context must be included, what outcome matters, what comparison would help and what approved evidence exists. Ask first the question whose answer would most change the decision, scope or evidence need. Pause and consult the stakeholder when two plausible readings would lead to different decisions, or when one request combines several decisions, populations or periods.

### WORKED EXAMPLE Worked Example: Scope Drift at OmniRetail

The weekend-ticket study begins focused. Stakeholders then ask to add pricing tiers, discount histories, shipping logistics and loyalty tiers without moving the deadline. If the team absorbs all of it, the study stalls and the staffing decision stays unresolved. The session teaches triage into **Deliver Now**, **Phase Later** or **Separate Charter**, with the timeline impact shown. In the slide example, adding shipping carrier delays would delay the staffing decision by eight weeks, so the analyst proposes the core comparison now and shipping as a Phase 2 study.

**WORKED EVIDENCE**      **Population:** Scope additions requested during the OmniRetail weekend support study (simulated case)

**Period:** During the active study, before the staffing decision

**Unit:** One requested addition

| Requested addition                                        | Triage category  | Reason                                                       |
|-----------------------------------------------------------|------------------|--------------------------------------------------------------|
| Core weekend support comparison                           | Deliver Now      | Directly informs the pending 450,000-dollar decision         |
| Shipping carrier delays                                   | Phase Later      | Adding it now delays the staffing decision by eight weeks    |
| Addition needing a different data source, owner or budget | Separate Charter | It is a different decision, not a larger version of this one |

**Interpretation:** Triage protects the decision timeline and records secondary questions instead of silently absorbing them.



#### LIMITS AND CONDITIONS

### Interpreting the Cost of Drift and Hidden Boundaries

When departments define retention differently, meetings stall over whose numbers are right instead of the strategic choice. When exclusions go unrecorded, readers cannot tell which cases the finding describes. The session's rule: log every excluded observation with an explicit rule, a reason and an exact record count, and place limitations in the executive summary and main recommendation rather than in footnotes. A fixed window, such as 30, 60 or 90 days after an event, gives every case the same observation period and keeps seasonal surges from distorting churn.



#### COMMON MISTAKE

### Common Misconception: Saying Yes to Every Addition Is Good Service

Accepting every addition feels helpful, but it delays the timeline, multiplies data complexity and leaves the core decision unaddressed. Refusing is also a mistake. The middle path is documented triage that shows the effect on time and data, with reference to the agreed project charter or a formal change request. Early formulas are not progress either: written before definitions are agreed, they must be redone.



## LIMITS AND CONDITIONS

### Limits of Scope Control

Scope control cannot guarantee that the boundaries chosen are the right ones, and a phased plan can leave an important question waiting. When a stakeholder insists on a boundary that would bias a comparison, the analyst may report subgroups separately or state that the framing cannot support the intended conclusion. If clarification does not arrive before a deadline, proceed only with clearly labeled provisional boundaries; the readiness classification in Section 4 may then be "clarification required."



## GUIDED PRACTICE

### Guided Practice: A Paired Boundary Audit

Audit your draft from 1.2.1.7, or a partner's. Task 1: Circle every word that two reasonable people could interpret differently. Task 2: For each, propose a bounded replacement or mark it as a clarification question. Task 3: State one included and one excluded context. Task 4: Write the single highest-value clarification question.

**Optional independent practice: my response**

*Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.*



## TRANSFER

### Changed-Case Transfer: A Biased Workforce Request

OmniRetail's Director of Customer Operations asks: "Prove that our remote support agents perform worse than our on-site team." The session's neutral frame covers forty-five full-time tier-1 agents over the trailing six months, excludes temporary contractors whose training and tenure differ, uses First-Contact Resolution (FCR, the share of tickets resolved in the first interaction) and 30-day repeat-ticket volume, and agrees in writing that only a gap of more than five percentage points would justify mandatory office relocation. Task 1: Rewrite the mandate without the word "prove." Task 2: Explain why contractors are excluded. Task 3: If the director objects, state the risk you would highlight and the session's alternative (report full-time staff and contractors separately).

**Optional independent practice: my response**

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.

## KEY TERMS Key terms

### Scope Drift:

Expansion of questions, cohorts or metrics without adjusting timeline or data budget.

### Premature Metric Coding:

Writing formulas before stakeholders agree on definitions.

### Concealed Limitation:

A known gap or exclusion that is omitted or buried, so readers assume universal findings.

### Clarification Ladder:

The ordered clarification questions, starting with what decision could change and who owns it.

### Scope Triage:

Sorting added requests into Deliver Now, Phase Later or Separate Charter.

## STUDY CHECK Put the learning into action

- Replace or flag vague words in every draft before calculation.
- Log every exclusion with a rule, a reason and a record count.
- Handle added requests with documented triage.
- Reflect on which framing choices came from the stakeholder and which you introduced.

## REFERENCE SUMMARY Reference summary

Use these concise reminders after reading the complete explanations above.

## CONCEPT The Components of a Bounded Question

### THREE TIERS

- Decision Context (stakeholder, decision, threshold); Evidence Scope (population, exclusions, horizon); Measurement Plan (outcome, reference point, verified source).

## CONCEPT Vague Versus Bounded

### VAGUE

- "Do weekend tickets hurt retention?"

### BOUNDED

- Adds a named population, an exclusion, a time horizon, a 60-day outcome window and a comparison group.

## ⚠️ COMMON MISTAKE Three Scoping Mistakes

### NAME THEM

- Scope drift, premature metric coding, concealed limitations.

### RESPOND

- Triage additions, agree definitions first, and state limitations in the summary.

## CONCEPT Clarifying Before Framing

### ASK IN ORDER

- What decision could change? Who owns it, and by when? What context, outcome, comparison and evidence matter?

### LABEL YOUR OWN CHOICES

- A boundary set without confirmation is an assumption.

## ✅ STUDY CHECK

### Study check: From a Managerial Concern to a Bounded Analytic Question

Use this ungraded check after reading. Close the explanation before responding, then reopen it to identify a specific correction.

1. Explain one central idea in your own words and connect it to a worked example above.
2. Reconstruct the example's inputs, sequence and interpretation; state what evidence supports each step.

3. Change one assumption or condition. Explain whether the same approach still applies.
4. Name a likely mistake and the check you would use to detect it.

### My response

*Write your explanation, evidence and changed-condition reasoning here.*

### MY READINESS

Choose one: **Can explain independently** / **Can explain with the reference**  
**/ Need to revisit.**

*Record the passage to revisit and the next action you will take.*

Carry one unresolved question into the next section.



### CONNECT TO THE LAB

### Connect this section to the lab

[Open Part E in the lab guide →](#)

#### Session lab reference

S01P (ungraded practice): MetroFresh Logistics: Are our order records ready to use as evidence?

#### Open the assigned task

Part E: Can we use these records as evidence, or should we pause?

#### Student lab file

`S01P_Student_Lab.html`, with `Session 01 - Analysis Practice Workbook.xlsx` and `Session 01 - SPSS Practice Syntax.sps`

#### Workbook location

'Synthesis'!A5

#### SPSS location

The Descriptives table from the DESCRIPTIVES command, as a source for cited numbers

#### Primary task

Write a 150 to 200 word evidence-sufficiency note that answers the Vice President's actual question: use these records as evidence, or pause?

#### Supporting task

Stay inside the request's scope. The email asks whether the records are ready to use as evidence, not whether to change service coverage.

### Excel task

Replace the placeholder in 'Synthesis'!A5 with a clear use-or-pause recommendation, at least three traceable numbers, what the records can describe, what they cannot establish and the main limitation.

### SPSS task

No new syntax. Confirm that each cited number also appears in your saved Descriptives output.

### Bring back to the discussion

Your use-or-pause recommendation in one sentence.

This session's practice is ungraded and nothing is submitted. Follow the practice guide's own setup steps, instructions and checklist. The Toolkit is a study reference and does not change those requirements.

### My connection

*Which idea from this section supports my lab decision, and what evidence will I check?*

[Back to contents](#)

## 1.3 · Testable Claims, Assumptions, and Evidence Boundaries

### IN CLASS

#### [Session slides](#)

Assumptions and limitations 21

MVR quality checklist 32

### PRACTISE IT IN THE LAB

- [Part D: Which records need a second look?](#) · 8 min

About 15 min read · PDF pages 29–42 · [Short version \(Concise\)](#)

### Learning goals

- Audit a proposed claim for testability and revise language that exceeds what the anticipated evidence could support.
- Distinguish an assumption to validate from a limitation to disclose.
- Use associational wording for observational records and separate statistical detectability from practical size.

## Testable Claims, Disconfirmation, and Practical Thresholds

### CONCEPT What a Testable Claim Means

A **provisional claim** states what the evidence might show, written before the evidence is examined. It is **testable** when observable evidence could count for or against it; the slides call this being **falsifiable**. "Customer service quality matters" is not testable: it has no measurable outcome, comparison or threshold, so no observation could challenge it. **Claim strength** cannot exceed the relevance, provenance, scope, timing or design of the evidence. A claim about one region and period cannot carry a conclusion about all regions and future periods.

## CONCEPT How a Claim Audit Works

The session's claim audit checks five things: scope, comparison, direction, time horizon and decision relevance, followed by a check of provenance and design. Flag words that expand causality (caused, produced, led to, resulted in) or certainty and scope (will, every, eliminate). Each flagged word needs design support or revision. Then write a **disconfirmation rule** before querying data by completing: "My claim would be challenged if..." Finally, sort the claim as supportable, needs qualification or unsupported under the anticipated design.

## WORKED EXAMPLE Worked Example: OmniRetail's Claim and Threshold

The OmniRetail claim reads: "Active monthly subscribers experiencing unresolved weekend support tickets exhibit a 60-day cancellation rate at least five percentage points higher than those with resolved weekday tickets." The five-point threshold is agreed with leadership before any query. The slides illustrate the disconfirmation rule: if the observed difference is only 1.2 percentage points, the team must acknowledge that weekend delays do not justify a 450,000-dollar budget. The claim-strength sort below assumes observed records, not a controlled experiment.

**WORKED EVIDENCE**      **Population:** Claims about observed records within a stated population and period

**Period:** Session 01, Section 3 claim-strength sort

**Unit:** One claim

| Claim                                                            | Strength under an observed-records design | Why                                                                        |
|------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------|
| Approved records document a difference within the stated period. | Supportable                               | Matches what observed records can show                                     |
| The program caused the difference.                               | Needs qualification or unsupported        | Records alone do not settle assignment, timing or alternative explanations |
| The program will improve outcomes for every customer.            | Unsupported                               | Claims certainty and universality beyond the stated scope                  |

**Interpretation:** Strength is set by the design and scope of the evidence, not by the confidence of the wording.

### **CONCEPT** Interpreting Statistical Versus Practical Significance

The threshold matters most in large datasets. In the session's illustration, across 2.4 million records, customers with unresolved weekend tickets show a 0.3-percentage-point higher cancellation rate, statistically significant at  $p < 0.001$ . **Statistical significance** means a difference is unlikely to be pure sampling noise; with millions of records, tiny differences qualify. **Practical significance** asks whether the difference is large enough to justify spending or policy change. The slides value the 0.3-point difference at about 24,000 dollars of saved annual revenue, against a 450,000-dollar cost. Statistical detectability is not economic value, which is why the threshold is agreed before results are seen.

### **COMMON MISTAKE** Common Misconception: Disconfirming the Sponsor Is a Failure

Under pressure, analysts may lower a threshold afterward, add conditions until the preferred answer appears, or present a tiny significant difference as if it justified the proposal. The session treats each as an integrity failure; a disconfirmation is reported transparently. Careful wording also cannot compensate for weak evidence: it can bound a claim but cannot repair unsuitable evidence. When asked to prove a preferred answer, reframe neutrally and state what evidence could count against it.

### **LIMITS AND CONDITIONS** Limits of Testability and Thresholds

A testable claim can still be tested on the wrong population. A threshold such as five points is a business agreement, not a statistical law; different costs could justify a different threshold. If a decision requires a causal conclusion that the planned evidence cannot give, request design information or say so plainly. If missing, prohibited or unsuitable evidence would make any result misleading, recommend no immediate analysis.

### **GUIDED PRACTICE** Guided Practice: Audit an Executive Claim

The session's checkpoint showed: "Hiring 20 weekend agents will eliminate the 14% increase in cancellations." Option A called it defensible because it proposes a concrete action. Option B called it flawed because it asserts absolute causality and guarantees outcomes. Task 1: Choose the defensible

evaluation and explain why. Task 2: Identify the words that carry the most risk. Task 4: Write a defensible reframe, then compare it with the session's model: because unresolved weekend tickets are associated with elevated cancellation, recommend a 90-day pilot of weekend support across three test regions to evaluate the effect on retention.

#### Optional independent practice: my response

*Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.*



### Changed-Case Transfer: "Customer Service Quality Matters"

A leadership deck states: "Customer service quality matters." Task 1: Explain why it is not testable. Task 2: Using the OmniRetail frame or the workforce frame from 1.2.2.8, rewrite it as a provisional claim with a population, period, outcome, comparison and direction. Task 3: Complete "My claim would be challenged if..."

#### Optional independent practice: my response

*Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.*



### Precommitted Decision Threshold Rule

$$\Delta = p_{\text{weekend, unresolved}} - p_{\text{weekday, resolved}}, \quad \text{supports the commitment only if } \Delta \geq \delta$$

#### SYMBOLS AND PRONUNCIATION

- $\Delta$  (say "delta"): the observed difference in 60-day cancellation rates, in percentage points.
- $p_{\text{weekend, unresolved}}$  and  $p_{\text{weekday, resolved}}$ : the 60-day cancellation rates after the two kinds of ticket.
- $\delta$  (say "small delta"): the minimum actionable difference agreed in advance; for OmniRetail,  $\delta = 5$ .

## READING THE NOTATION

- $\geq$  (say "greater than or equal to"): the observed difference must reach the agreed threshold.

## INPUTS AND ASSUMPTIONS

A bounded question, an agreed reference group, and a threshold written down before any query.

## Worked steps

- 1 Step 1: Agree and date  $\delta$  with the decision owner before any query.
- 2 Step 2: Write the rule: if  $\Delta < \delta$ , the evidence does not justify the commitment.
- 3 Step 3: After later analysis estimates  $\Delta$ , compare without changing  $\delta$  or the reference group.

## WORKED RESULT

With  $\delta = 5$ :  $\Delta = 1.2$  gives  $1.2 < 5$ , so the budget is not justified.  $\Delta = 0.3$  gives  $0.3 < 5$  even if statistically significant; the slides value it at about 24,000 dollars a year against a 450,000-dollar cost.

## WHAT THE RESULT MEANS

The rule separates "Is there a difference?" from "Is it big enough to act on?" Meeting the threshold does not prove cause and does not make the decision.

## WHEN TO USE IT

Whenever evidence will justify a resource commitment, especially with very large datasets.

## LIMITS AND CONDITIONS

The threshold is a business agreement. Estimating  $\Delta$  and its uncertainty belongs to later sessions.

## HOW TO CHECK THE REFERENCE

The threshold's recorded date is earlier than the first data query, and the reported reference group matches the frame.



## KEY TERMS

### Key terms

#### Testable Claim:

A statement specific enough that observable evidence could count for or against it.

#### Disconfirmation Rule:

A statement, written before data are examined, of what would count against the claim.

### Claim Strength:

How far a claim reaches in scope, certainty and causality; it cannot exceed the evidence.

### Statistical Significance:

An indication that a difference is unlikely to be sampling noise; not a measure of business value.

### Practical Significance:

Whether a difference is large enough to justify committing capital, staff or policy.

## ✔ STUDY CHECK Put the learning into action

- Turn each bounded question into a provisional claim and complete "My claim would be challenged if..."
- Agree and date a practical threshold before querying data.
- Sort claims as supportable, needs qualification or unsupported, and revise the last two.
- Reflect on what evidence would change your own preferred interpretation.

## Assumptions, Limitations, and Causal Language

### 🔧 LIMITS AND CONDITIONS What Assumptions and Limitations Mean

An **assumption** is a condition accepted provisionally so work can proceed; it must be audited. Assuming that OmniRetail's ticket creation timestamps record when customers experienced friction is an assumption that can be checked against server error logs. A **limitation** is a structural boundary on interpretation or use that the available data or design cannot remove. Restricting the study to United States consumer subscribers is a limitation: the findings cannot justify support policy for enterprise business-to-business accounts. In the podcast's words, you audit assumptions in a log; you transparently disclose limitations. A limitation narrows a valid conclusion without making it worthless.



## LIMITS AND CONDITIONS

### How the Assumption and Limitation Log Works

Use two columns: **Assumption to validate** and **Limitation to disclose**. Pair each assumption with a validation method and each limitation with a boundary statement. Treat unverified recording consistency as an assumption, not a fact. Limitations cannot be fixed with more calculation, because they come from the population, period or observational design. In the practice, the Quality Log plays this role, with an action (keep the value) and a remaining question for each flagged record.



## WORKED EXAMPLE

### Worked Example: Revising a Causal Claim About Onboarding

The instructor's guide revises the claim "The new onboarding program caused customers to renew." The planned evidence is observed participation records, which show who took part and who renewed, not why, so "caused" is unsupported. Each part of the revision is shown below.

**WORKED EVIDENCE**      **Population:** Customers in the approved onboarding records (instructor's worked example)

**Period:** The approved period named in the frame

**Unit:** One log or claim entry

| Element                   | Entry                                                                                                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Original claim            | The new onboarding program caused customers to renew.                                                                                                                              |
| Revised claim             | Within the approved population and period, documented renewal was more common among recorded participants than recorded nonparticipants; these records alone do not establish why. |
| Assumption to validate    | Participation and renewal statuses were recorded consistently; check against the policy documentation.                                                                             |
| Limitation to disclose    | One region and one period may not represent other settings or future periods.                                                                                                      |
| Disconfirming observation | The scoped records show no such pattern, or the opposite pattern.                                                                                                                  |

**Interpretation:** The revision keeps what the records can show and removes what they cannot.

## CONCEPT Interpreting Observational Patterns: How Confounding Misleads

Most business records, including OmniRetail's tickets, are **observational**: they record what happened without the analyst controlling who received what. They show associations and sequences, not causes. The slides' example: subscribers who joined with a discount code show 35 percent higher cancellation, and an executive concludes discounts cause churn. The likely driver is **price sensitivity**, which leads customers both to seek coupons and to cancel when full billing begins. It is a **confounder**, a third variable affecting both the supposed cause and the outcome. Removing discounts would not make these customers loyal. List at least two alternative explanations before presenting an observational recommendation.

## COMMON MISTAKE Common Misconception: Significant Results Justify Causal Claims

Statistical significance in observational data, even in a regression model, indicates association after accounting for the included variables, not cause. Significance measures sampling variability, not whether the causal story is identified. Swapping "caused" for "impacted" or "drove" still implies cause; the session recommends "was associated with," "co-occurred with" or "differed between groups."

## LIMITS AND CONDITIONS Limits of Observational Evidence and the Role of a Pilot

Causal language needs a supporting design, such as a randomized experiment or a strong quasi-experimental design. When records show only an association, the session's middle path is a **controlled operational pilot**. Instead of funding company-wide weekend support, the Vice President could fund a 90-day pilot in three representative regions, keep the other regions as a control group, and track First-Contact Resolution, repeat contacts and 60-day cancellation in both. Its window, sample size and method are fixed before launch.

## GUIDED PRACTICE Guided Practice: Build an Assumption and Limitation Log

Use the OmniRetail frame. Task 1: Draw two columns, "Assumption to validate" and "Limitation to disclose." Task 2: Place each statement and add its action: timestamps record when customers experienced friction; the study covers only United States consumer subscribers; cancellation records are synchronized across web and mobile billing systems; the evidence comes from observational records. Task 3: Explain why timestamp accuracy is an assumption, using the session's reasons (batch processing queues, system outages, time zone logging mismatches).

### Optional independent practice: my response

*Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.*



TRANSFER

### Changed-Case Transfer: Remote Versus On-Site Agents

Return to the workforce frame in 1.2.2.8. Suppose a later comparison reports a difference between remote and on-site agents. Task 1: Write the finding in associational language. Task 2: Name two alternative explanations that could produce a difference without location being the cause; the reason for excluding contractors (different onboarding and shift patterns) is a starting point. Task 3: Write one assumption to validate and one limitation to disclose.

### Optional independent practice: my response

*Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.*



CONCEPT

### Claim-Language Reference for Observational Evidence

Reference for reading and optional independent practice.

Flag these words  
caused, produced  
led to, resulted in  
drove, impacted  
will, eliminate  
every, all customers  
prove (in a mandate)

Why they need design support  
assert cause and effect  
assert cause and effect  
imply cause without saying so  
promise a guaranteed outcome  
assert universal scope  
presumes the conclusion

Safer observational wording  
was associated with  
co-occurred with  
differed between groups  
in the stated population and period  
among [named population]  
evaluate whether

### DEFINITIONS

- Causal and certainty words: Words asserting that one factor produced an outcome, or that an outcome will happen.
- Observational wording: Phrasing that reports what records show without claiming why.

## INPUTS AND ASSUMPTIONS

A draft claim or title, and knowledge of whether the evidence is observed records or a controlled test.

### Worked steps

- 1 Step 1: Underline every left-column word in your draft.
- 2 Step 2: Keep it only if the design supports it; otherwise replace it and add population and period.
- 3 Step 3: Add two alternative explanations and one limitation beside the revised claim.

## WORKED RESULT

"Hiring weekend agents will eliminate the increase in cancellations" becomes "Unresolved weekend tickets were associated with elevated cancellation in the stated population and period; a 90-day pilot in three regions could evaluate the effect on retention."

## WHAT THE RESULT MEANS

The useful observation and the proposed action remain; the guarantee and the unsupported cause are removed.

## WHEN TO USE IT

Before sending any finding, note or slide title based on observational records.

## LIMITS AND CONDITIONS

Rewording cannot repair unsuitable evidence. If the decision needs a causal answer, request design information.

## HOW TO CHECK THE REFERENCE

Ask a partner to underline any remaining word that expands scope, certainty or causality.



## KEY TERMS

### Key terms

#### Assumption:

A condition accepted provisionally so work can proceed, which must be audited.

#### Limitation:

A structural boundary on interpretation or use that the available data or design cannot remove.

#### Observational Data:

Records of what happened without the analyst controlling who received which condition.

#### Confounder:

A third variable that influences both the supposed cause and the outcome.

### Associational Language:

Wording such as "was associated with" that reports a pattern without claiming its cause.

### ✓ STUDY CHECK Put the learning into action

- Keep a two-column log pairing assumptions with validation and limitations with boundary statements.
- Replace causal and certainty words when the design is observational.
- List two alternative explanations before presenting an observational finding.
- Reflect on whether a limitation is being hidden because it complicates the recommendation.

### 📌 REFERENCE SUMMARY Reference summary

Use these concise reminders after reading the complete explanations above.

### CONCEPT Testable Claims

#### MAKE IT CHALLENGEABLE

- An observable result must be able to count against the claim. Write "My claim would be challenged if..." before looking at data.

### CONCEPT Significant Is Not the Same as Worth Acting On

#### DETECTABLE VERSUS VALUABLE

- A 0.3-point difference in 2.4 million records can be significant yet worth about 24,000 dollars a year against a 450,000-dollar cost. The agreed bar is five points.



## LIMITS AND CONDITIONS

### Assumption or Limitation?

#### VALIDATE OR DISCLOSE

- Timestamp accuracy: an assumption to check. Consumer-only scope: a limitation to state prominently.

## CONCEPT

### Causal Language

#### MATCH WORDS TO DESIGN

- Observed records show associations. Watch for confounders such as price sensitivity, say "was associated with," and propose a controlled pilot when cause matters.



## STUDY CHECK

### Study check: Testable Claims, Assumptions, and Evidence Boundaries

Use this ungraded check after reading. Close the explanation before responding, then reopen it to identify a specific correction.

1. Explain one central idea in your own words and connect it to a worked example above.
2. Reconstruct the example's inputs, sequence and interpretation; state what evidence supports each step.
3. Change one assumption or condition. Explain whether the same approach still applies.
4. Name a likely mistake and the check you would use to detect it.

#### My response

*Write your explanation, evidence and changed-condition reasoning here.*

#### MY READINESS

Choose one: **Can explain independently** / **Can explain with the reference** / **Need to revisit.**

*Record the passage to revisit and the next action you will take.*

Carry one unresolved question into the next section.

[Open Part D in the lab guide →](#)

### Session lab reference

S01P (ungraded practice): MetroFresh Logistics: Are our order records ready to use as evidence?

### Open the assigned task

Part D: Which records need a second look?

### Student lab file

S01P\_Student\_Lab.html, with Session 01 - Analysis Practice Workbook.xlsx and Session 01 - SPSS Practice Syntax.sps

### Workbook location

'Quality Log'!A6:F13

### SPSS location

No new block; use the valid N values in your saved Descriptives output

### Primary task

Log each blank field, the largest delivery time and the items-rule result, keeping every source value as recorded.

### Supporting task

Separate what you can check later (an assumption, such as whether a delivery time was recorded correctly) from what you must disclose (a limitation, such as records covering only four weeks). Carry the biggest issue into 'Data Audit'!B11 and your Part E note.

### Excel task

Use Data > Filter on the Data sheet to find each order\_id, then clear the filter. Record order\_id, field, recorded value, reason, action (keep the value) and remaining question. If nothing breaks the items rule, write one row saying the check passed.

### SPSS task

Compare each variable's valid N with the 120 cases in Data View; the difference is the number of blanks to log.

### Bring back to the discussion

One record you flagged and the question that remains.

This session's practice is ungraded and nothing is submitted. Follow the practice guide's own setup steps, instructions and checklist. The Toolkit is a study reference and does not change those requirements.

### My connection

*Which idea from this section supports my lab decision, and what evidence will I check?*

[Back to contents](#)

## 1.4 · Minimum Viable Readiness in Excel and SPSS

### IN CLASS

#### Session slides

Excel and SPSS roles 29

Spreadsheet modeling standards 30

Cross-tool reproducibility 49

SPSS workspace architecture 31

Readiness trace 33

### PRACTISE IT IN THE LAB

- Part B: What do delivery time, order value and spoilage look like? · 12 min
- Part C: Do Excel and SPSS give the same answer? · 10 min · SPSS Descriptive Statistics
- Part F: Where does each number in the note come from? · 6 min

About 16 min read · PDF pages 43–56 · [Short version \(Concise\)](#)

### Learning goals

- Demonstrate minimum readiness in Excel and SPSS: open or import provided files, execute provided steps, inspect results, save and reopen artifacts, and record provenance.
- Explain why cross-tool agreement confirms consistent processing but not accurate records.
- Classify a request as ready to analyze, clarification required, evidence blocked or tool blocked.

## Excel and SPSS Readiness Traces

### CONCEPT What Minimum Viable Readiness Means

**Minimum Viable Readiness (MVR)** is observable proof that your workflow functions, not just that an application launches. The slides give three pillars. **Source protection:** raw data stay untouched and all work happens in a named working copy. **Dynamic execution:** changing an Excel input updates linked outputs, and saved SPSS syntax runs cleanly. **Evidence verification:** close the applications, reopen the saved files and confirm the results still appear without errors. Desktop Excel 365 and IBM SPSS Statistics are required; Python and R are optional.

## CONCEPT How Each Tool Organizes Work

Excel keeps data, formulas and results in one grid, so structure must be designed. The **input layer** holds assumptions and parameters in labeled cells; the **logic layer** holds formulas that refer only to those cells; the **output layer** gathers the decision metrics. Typed numbers inside formulas (buried constants) hide their meaning. SPSS separates three windows and files. The **Data Editor** (.sav) has Data View (records in rows) and Variable View (names, labels, measurement levels, missing-value codes). The **Syntax Editor** (.sps) holds commands. The **Output Viewer** (.spv) shows tables, logs and warnings. Saving one does not save the others.

## WORKED EXAMPLE Worked Example: The Excel and SPSS Readiness Traces

In the Excel trace, open the working copy, confirm its name, find the input and output areas, make a reversible change, watch the linked output respond, restore the value and save. The slides change an hourly support wage from 20 to 25 dollars and back; the instructor's guide switches a selector from Base to Check and back. No formula is created. In the SPSS trace, open data and syntax separately, run only the named block, check the log in the Output Viewer, save each artifact, export the requested output, close everything and reopen from the locations in your readiness log.

**WORKED EVIDENCE**      **Population:** Files created during the Session 01 readiness demonstration

**Period:** The Tool Readiness section of the slide deck

**Unit:** One saved artifact

| Artifact                                           | Evidence it provides                                                         |
|----------------------------------------------------|------------------------------------------------------------------------------|
| Untouched source file                              | The baseline can be restored and checked                                     |
| Named working copy (.xlsx)                         | The reversible change recalculated and was restored                          |
| Data (.sav), syntax (.sps) and output (.spv) files | Records, repeatable commands and error-free results, each saved separately   |
| Readiness log                                      | Source, file name, action, observed result, saved location, unresolved issue |

**Interpretation:** Readiness is proven by saved, reopened artifacts and a log someone else can follow, not by a screenshot.

### **CONCEPT** Interpreting What the Traces Show

The Excel change asks whether outputs depend on inputs. If nothing updates, check for pasted values or a calculation setting that is not Automatic. The SPSS run asks whether saved commands execute cleanly; read the log and warnings at the top of the output first. The session's two most common syntax faults are a missing period at the end of a command and a variable name that does not match Variable View. The reopen test asks whether the work persisted.

### **COMMON MISTAKE** Common Misconception: A Screenshot or a Launch Proves Readiness

Launching Excel or SPSS shows only that the application opens. A screenshot captures a static picture; it cannot show that formulas are dynamic, syntax is repeatable or files reopen. Clicking OK in an SPSS dialog is not the same as saved syntax: the session advises clicking **Paste**, which writes the command into the Syntax Editor so it can be saved, audited and rerun.

### **LIMITS AND CONDITIONS** Limits of a Readiness Trace

A trace verifies workflow, not content. A successful import confirms access, not correctness or provenance; a clean run confirms execution, not accurate values. An accessible keyboard or nonvisual equivalent counts if it preserves the same evidence. If artifacts cannot be created, saved, reopened or inspected, the classification is "tool blocked," recorded with the exact error message text.

### **GUIDED PRACTICE** Guided Practice: Run the Practice Import and Confirm 120 Cases

This is the S01P "Before class" setup. Task 1: Put the practice workbook and `Session 01 - SPSS Practice Syntax.sps` in one folder, because SPSS reads the workbook's Data sheet. Task 2: Open the workbook, read Start Here, and use File > Save As with the pattern `Lab01_FirstName_LastName_NetID_Analysis.xlsx` (NetID is the part before @nyu.edu); leave no

template brackets. Task 3: Run the GET DATA block with Run > Selection; if SPSS cannot find the file, put its full path after /FILE=. Task 5: Write a readiness log entry. You are ready when SPSS shows 120 cases and your copy is saved under your own name.

### Optional independent practice: my response

*Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.*



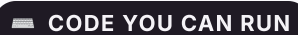
TRANSFER

## Changed-Case Transfer: Missing Leading Zeros

An analyst imports OmniRetail's customer table and identifier 00458921 becomes 458921. Identifiers are labels, not quantities, so ticket and billing records no longer match. Task 1: Explain why account numbers, zip codes and phone numbers belong in text form. Task 2: Explain why a custom display format such as 00000000 does not fix it (it changes only the display; the stored value stays truncated). Task 3: Describe the session's upstream fixes: in Excel, Power Query (Data > From Text/CSV, Transform Data) with the column set to Text before loading; in SPSS, a String variable with enough width.

### Optional independent practice: my response

*Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.*



CODE YOU CAN RUN

## SPSS Practice Syntax for the Shared Workbook

Code reference for reading and optional independent practice.

```
PRESERVE.
SET DECIMAL DOT.

GET DATA
  /TYPE=XLSX
  /FILE='Session 01 - Analysis Practice Workbook.xlsx'
  /SHEET=name 'Data'
  /CELLRANGE=FULL
  /READNAMES=ON.

DATASET NAME MetroFreshLab01 WINDOW=FRONT.
```

#### FORMATS

```
items_ordered items_unfulfilled delivery_minutes spoilage_units customer_rating (F8.0)
order_value (F10.2).
```

#### VARIABLE LABELS

```
delivery_minutes 'Elapsed minutes from dispatch to delivery'
order_value 'Recorded order value in U.S. dollars'
spoilage_units 'Units written off as spoiled'.
```

#### VARIABLE LEVEL

```
order_id customer_id fulfillment_hub product_category (NOMINAL)
supplier_tier customer_rating (ORDINAL)
items_ordered items_unfulfilled delivery_minutes order_value spoilage_units (SCALE).
```

EXECUTE.

```
DESCRIPTIVES VARIABLES=delivery_minutes order_value spoilage_units
/STATISTICS=MEAN STDDEV MIN MAX.
```

RESTORE.

## DEFINITIONS

- **GET DATA:** Reads the workbook's Data sheet, taking variable names from the first row.
- **DATASET NAME:** Names the open dataset MetroFreshLab01.
- **FORMATS and VARIABLE LABELS:** Set display width and readable descriptions; stored values do not change.
- **VARIABLE LEVEL:** Sets each field as nominal, ordinal or scale.
- **DESCRIPTIVES:** Reports valid N, mean, sample standard deviation, minimum and maximum.

## INPUTS AND ASSUMPTIONS

The syntax and workbook in one folder; every command ends with a period. The supplied file adds comment lines (starting with \*) that SPSS ignores.

## Worked steps

- 1 Step 1: Check the /FILE= path, run GET DATA and confirm 120 cases.

- 2 Step 2: Run the whole file with Run > All and inspect Variable View.
- 3 Step 3: Read the log and Descriptives table, save the .spv under your own name, close and reopen it.

|                                   |                                                                                                                         |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>WORKED RESULT</b>              | 120 cases; valid N of 118, 119 and 120 for the three variables; fifteen shared numbers to compare with Excel in Part C. |
| <b>WHAT THE RESULT MEANS</b>      | Saved syntax lets anyone rerun the same steps on the same file, which is what makes the work inspectable.               |
| <b>WHEN TO USE IT</b>             | Whenever a step must be repeatable rather than dependent on remembered clicks.                                          |
| <b>LIMITS AND CONDITIONS</b>      | Files in different folders break GET DATA. A clean run does not confirm accurate values.                                |
| <b>HOW TO CHECK THE REFERENCE</b> | 120 cases in Data View, no error text in the log, and valid N matching Excel's COUNT results.                           |



#### KEY TERMS

### Key terms

#### Minimum Viable Readiness (MVR):

Observable proof of source protection, dynamic execution and evidence verification.

#### Working Copy:

A named copy in which all changes are made, leaving the source untouched.

#### Reversible Recalculation Test:

Changing an input, confirming outputs update, then restoring the value.

#### Data Editor, Syntax Editor, Output Viewer:

The three SPSS workspaces for data (.sav), commands (.sps) and results (.spv).

#### Readiness Log:

A record of source, file name, action, result, saved location and unresolved issue.



#### STUDY CHECK

### Put the learning into action

- Save a named working copy before any change.

- Run a reversible recalculation test before trusting an inherited workbook.
- Click Paste in SPSS dialogs so every procedure is saved as syntax.
- Reflect on the least reliable step in your file workflow and name a recovery action.

## Inspectable Evidence, Cross-Tool Parity, and the Readiness Decision

### CONCEPT What Inspectable Evidence and Cross-Tool Parity Mean

An **inspectable evidence package** contains the untouched source, a named working copy, the saved output, the provided instructions or syntax, and a short provenance note. In the practice, every number in your note must also point to a workbook cell and an SPSS output. **Cross-tool parity** means two tools reading the same data produce the same numbers. When Excel and SPSS disagree, the cause is almost always a human difference in filtering, missing-value handling, data typing or formula choice, not an arithmetic bug. The practice therefore uses one Data sheet for every tool and requires agreement within 0.01.

### CONCEPT How Parity Checks Catch Rule Differences

Reconcile counts before comparing any summary. In Excel, COUNT counts numbers only and AVERAGE skips blanks; SPSS uses explicit system-missing and user-defined missing codes. For spread, Excel's STDEV.S divides by  $n$  minus 1, as SPSS does; STDEV.P divides by  $n$ . The practice's parity sheet computes the absolute difference in column E, holds the 0.01 tolerance in column F and shows PASS or CHECK in column G. Column C must be a formula pointing at your Excel result; typing the SPSS number there would compare SPSS with itself.

### WORKED EXAMPLE Worked Example: MetroFresh Parity and the Row Check

The S01P workbook holds 120 MetroFresh orders from August 1 to 28, 2026, placed by 76 distinct customers. Its Descriptive Statistics sheet summarizes three fields; the check values below come from the practice guide. For delivery\_minutes, the median is 43.50 and the range is 118. The row check shows nothing was lost:  $118 + 2$ ,  $119 + 1$  and  $120 + 0$  each equal 120. The divisor matters: for delivery time, STDEV.S gives 14.17 but STDEV.P gives 14.11, a gap of 0.06 that fails the 0.01 tolerance.

## WORKED EVIDENCE

**Population:** MetroFresh Logistics order records in the S01P practice workbook (teaching dataset)

**Period:** August 1 to August 28, 2026 (four weeks)

**Unit:** One order; values in minutes, US dollars and units

| Variable          | Total N | Valid n | Missing | Mean   | Min   | Max    | Sample SD |
|-------------------|---------|---------|---------|--------|-------|--------|-----------|
| delivery_minutes  | 120     | 118     | 2       | 43.67  | 24.00 | 142.00 | 14.17     |
| order_value (USD) | 120     | 119     | 1       | 142.07 | 43.05 | 279.74 | 57.27     |
| spoilage_units    | 120     | 120     | 0       | 0.65   | 0.00  | 4.00   | 1.29      |

**Interpretation:** Valid n, mean, minimum, maximum and sample SD for three variables are the fifteen numbers Excel and SPSS must match. They describe the records; they do not say whether the records suit the Vice President's purpose.

## CONCEPT

### Interpreting PASS and the Readiness Classification

Fifteen PASS results mean two tools computed the same numbers under the same rules. They do not prove the records are accurate: a wrong value in the Data sheet gives the same wrong mean in both. The session closes with four **readiness classifications**. **Ready to analyze:** the question is bounded, evidence is feasible and permitted, and the workflow functions. **Clarification required:** intent, decision use, scope or acceptable claim strength is still ambiguous. **Evidence blocked:** needed evidence is missing, prohibited or unsuitable for the claim. **Tool blocked:** the workflow cannot produce persistent, inspectable artifacts. A documented pause is valid professional work.



## COMMON MISTAKE

### Common Misconception: A File That Opens Is the Right File

The session plants a discrepancy: the instruction sheet names version 1, but the supplied file shows version 2. The right response is to stop, record both names, preserve the files and ask for confirmation, not to assume version 2 is correct because it opens. The practice adds two related rules. Never replace blanks with zero; if the minimum delivery time shows 0, a blank was changed and you should restart from a clean copy. Never cite a number worked out by hand; every number in the note needs a row in the Memo Evidence Map.



## LIMITS AND CONDITIONS

### Limits of Parity and Readiness Evidence

Parity means little if the tools read different files. It does not validate the Data sheet or show that the records suit the decision, and tool readiness cannot compensate for a weak question. In the practice, the records describe four weeks of orders; they cannot establish why any delivery took longer, and the 142-minute delivery time is a value to check, not delete. Optional Python and R scripts are reference checks only.



## GUIDED PRACTICE

### Guided Practice: Classify the Flexible Scheduling Request

A regional operations director asks analysts to use existing files to prove that flexible scheduling will improve service before a pilot is approved. Task 1: Silently complete a framing canvas: stakeholder, decision, context, included and excluded scope, time horizon, outcome, reference point, provisional claim, evidence need, assumptions, validation step, limitations and next action. Task 2: A partner gives one keep, one change and one verify comment. Task 3: Revise one element. Task 4: Classify the case with a one-sentence justification. Use the session's guidance: clarification required if locations, period, outcome or evidence standard are unspecified; evidence blocked if causal proof is required but the approved evidence cannot support it; tool blocked only when evidence exists but inspectable artifacts cannot be created.

#### Optional independent practice: my response

*Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.*



## TRANSFER

### Changed-Case Transfer: A Standard Deviation Row Shows CHECK

Fourteen parity rows show PASS, but the delivery\_minutes standard deviation row shows CHECK: Excel reports 14.11 and SPSS 14.17. Task 1: Identify the likely cause before touching data. Task 2: State the practice guide's fix (change column K of Descriptive Statistics from STDEV.P to STDEV.S). Task 3: Change the condition: a valid count row shows CHECK instead. What does that suggest (the tools read different data), and what would you verify first?

#### Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.

## CONCEPT Parity Rule and Row Check

$$| \text{Excel} - \text{SPSS} | \leq 0.01 \Rightarrow \text{PASS}, \quad n + (N - n) = N = 120$$

### SYMBOLS AND PRONUNCIATION

- $| \cdot |$  (say "absolute value of"): the size of a difference, ignoring its sign.
- $N$  (say "capital en"): total records, 120 in the practice.
- $n$  (say "en"): valid numeric values for one variable;  $N - n$  is the missing count.
- $\Rightarrow$  (say "implies"): if the left side holds, the right side follows.

### READING THE NOTATION

- The compared summaries include the sample standard deviation  $s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$ , where  $x_i$  is one valid value and  $\bar{x}$  the mean. Later sessions explain it.

### INPUTS AND ASSUMPTIONS

Excel formulas pointing at the Data table, SPSS values from the saved Descriptives output, and one shared Data sheet.

### Worked steps

- 1 Step 1: Confirm valid  $n$  plus missing equals 120 for each variable.
- 2 Step 2: Link Excel results by formula; type SPSS values with enough decimals.
- 3 Step 3: Investigate any CHECK through counts, filters, blanks and divisors before changing anything.

### WORKED RESULT

Delivery time:  $118 + 2 = 120$ . Sample SD with STDEV.S:  $| 14.17 - 14.17 | = 0$ , PASS. With STDEV.P:  $| 14.11 - 14.17 | = 0.06 > 0.01$ , CHECK.

### WHAT THE RESULT MEANS

The row check shows no record was lost or added; parity shows the tools agree. Neither shows the values are correct.

### WHEN TO USE IT

Before citing any number produced in more than one tool.

### LIMITS AND CONDITIONS

The 0.01 tolerance suits two-decimal results; it is a practice convention.

### HOW TO CHECK THE REFERENCE

All fifteen rows in 'Excel-SPSS Parity'!G6:G20 show PASS, and column C holds formulas.



### KEY TERMS

### Key terms

#### Inspectable Evidence Package:

Source, working copy, output, instructions or syntax, and a provenance note.

#### Cross-Tool Parity:

Agreement of two tools' results on the same data under the same rules, within a tolerance.

#### Valid N and Missing N:

Usable numeric values for a variable, and total records minus that count.

#### Readiness Classification:

Ready to analyze, clarification required, evidence blocked or tool blocked.

#### Evidence Map:

A table linking every number in a note to its workbook cell and SPSS output.



### STUDY CHECK

### Put the learning into action

- Reconcile counts before comparing summaries across tools.
- Investigate every CHECK by examining rules, not retyping values.
- Map every cited number to a workbook cell and an SPSS output.
- Reflect on whether another analyst could reproduce your work from your artifacts and log alone.



### REFERENCE SUMMARY

### Reference summary

Use these concise reminders after reading the complete explanations above.

## CONCEPT Minimum Viable Readiness

### THREE PILLARS AND THEIR PROOF

- Source protection, dynamic execution and evidence verification, proven by saved, reopened artifacts and a log, not a screenshot.

## CONCEPT Excel and SPSS Structures

### EXCEL AND SPSS

- Excel: inputs, logic and outputs kept separate, no buried constants. SPSS: .sav, .sps and .spv saved separately; click Paste, end commands with periods.

## CONCEPT Parity in the Practice

### SAME SOURCE, SAME RULES

- One Data sheet; COUNT for valid N; blanks left blank; STDEV.S. Fifteen PASS results show consistency, not accuracy.

## CONCEPT The Readiness Decision

### FOUR CLASSIFICATIONS

- Ready to analyze, clarification required, evidence blocked, tool blocked. Justify with visible evidence; a pause is valid work.

## ✔ STUDY CHECK Study check: Minimum Viable Readiness in Excel and SPSS

Use this ungraded check after reading. Close the explanation before responding, then reopen it to identify a specific correction.

1. Explain one central idea in your own words and connect it to a worked example above.
2. Reconstruct the example's inputs, sequence and interpretation; state what evidence supports each step.
3. Change one assumption or condition. Explain whether the same approach still applies.
4. Name a likely mistake and the check you would use to detect it.

**My response**

*Write your explanation, evidence and changed-condition reasoning here.*

### MY READINESS

Choose one: **Can explain independently** / **Can explain with the reference**  
**/ Need to revisit.**

*Record the passage to revisit and the next action you will take.*

Use your notes in the session readiness check below.



**CONNECT TO THE LAB**

**Connect this section to the lab**

**Open Part B in the lab guide →**

### Session lab reference

S01P (ungraded practice): MetroFresh Logistics: Are our order records ready to use as evidence?

### Open the assigned task

Part B (descriptive summaries), Part C (Excel and SPSS parity) and Part F (evidence map)

### Student lab file

S01P\_Student\_Lab.html, with Session 01 - Analysis Practice Workbook.xlsx and Session 01 - SPSS Practice Syntax.sps

### Workbook location

'Descriptive Statistics'!C6:K8; 'Excel-SPSS Parity'!C6:D20; 'Memo Evidence Map'!A6:G13

### SPSS location

The full practice syntax, run with Run > All, ending in DESCRIPTIVES

### Primary task

Build the Descriptive Statistics rows with formulas and reach PASS on all fifteen parity rows.

### Supporting task

Map every number in your Part E note to a cell and an SPSS output; type Yes only after reopening both files.

### Excel task

Enter the row 6 formulas, fill down, change the bracketed names, then link parity column C to your results by formula.

### SPSS task

Run the syntax, confirm 120 rows, save the Output Viewer under your own name, and type its N, Mean, Minimum, Maximum and Std. Deviation values into parity column D.

### **Bring back to the discussion**

The valid N and mean for delivery time, and how many of the 15 checks passed.

This session's practice is ungraded and nothing is submitted. Follow the practice guide's own setup steps, instructions and checklist. The Toolkit is a study reference and does not change those requirements.

### **My connection**

*Which idea from this section supports my lab decision, and what evidence will I check?*

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